

PROJECT ZENITH: CELESTIAL EYE

Space Intelligence Dashboard & Real-Time Astronomy Tracker

1. Website Functionality and Unique Features

Project Zenith (Celestial Eye) is a premium, real-time WebGL space intelligence application that allows users to track satellites, visualize celestial bodies, observe orbital paths, and calculate local sky viewing suitability. Built with a sleek dark-mode glassmorphic interface, it combines astronomical calculations with real-time telemetry.

Key Highlight Features:

- **Geocoded Search & Coordinate Autocomplete:** Users can type any city, country, or raw coordinates (e.g. 17.40, 78.37). The dashboard debounces input and queries OpenStreetMap Nominatim API in the background to provide instant location matching, flying the globe camera seamlessly to the target.
- **Cache-First GPS Tracking Fallback:** An optimistic tracking system resolves coordinates instantaneously. Next-click locks resolve in <10ms by leveraging local storage and standardizing maximumAge browser caching rules.
- **3D Interactive Globe:** Renders dynamic night-earth maps, city markers, real-time day/night shadow cycle, live ISS positions, and coordinates selection telemetry pulses.
- **SGP4 Live Satellite Propagation:** Runs real-time orbital calculations locally in the client browser using two-line element (TLE) datasets, showing trajectories and active radar scans.
- **3D Solar System View Portal:** Interactive canvas detailing exact planetary orbits, orbital rings, and asteroid belts, equipped with a fluid responsive fullscreen viewing mode.
- **Dynamic AI Space Guide:** Real-time observation advisor generating dynamic natural language sky summaries based on current coordinates, cloud covers, and planet heights.
- **Cosmic Event Predictor:** Real-time timers tracking upcoming cosmic events, planetary alignments, eclipses, and meteor showers relative to the active coordinate meridian.

2. Installation and Setup Instructions

Follow these steps to run the Project Zenith source code locally on your development system:

Prerequisites: Ensure that Node.js (version 18.0 or later) and npm are installed.

Step 1: Clone the Repository

```
git clone https://github.com/harshith7002/Project-Zenith.git
cd Project-Zenith
```

Step 2: Install Package Dependencies

```
npm install
```

Step 3: Run the Local Development Server

```
npm run dev
```

The local server will run on **http://localhost:3000**. Open this address in your web browser.

Step 4: Build for Production Production Deployment

```
npm run build
npm run start
```

3. Core Project Dependencies

The application runs on a modern reactive stack utilizing WebGL rendering layers and orbit propagation math engines:

Dependency Layer	Libraries/Tools Used	Purpose
Application Core	React 19, Next.js 15, TypeScript	Reactive frontend state rendering, Next.js server-side static generation.
3D Visualization	Three.js, @react-three/fiber, @react-three/drei	Rendering 3D orbital controls, interactive planets, solar system orbits, and stars.
Interactive Globe	react-globe.gl, three-globe	3D day/night night-lights earth globes mapping satellite paths.
Orbit Maths	satellite.js, astronomy-engine	TLE SGP4 satellite propagation calculations and planetary horizontal coordinate positions.
Styles & Motion	TailwindCSS, Vanilla CSS, framer-motion	Custom glassmorphic card overlays, responsive telemetry pulse animations.